

Roll Number:C24120

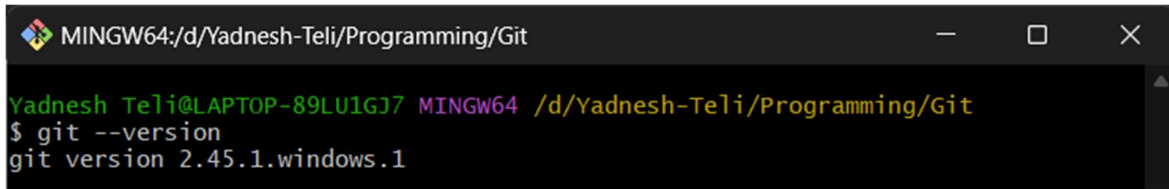
MCAL23 DevOps Lab INDEX				
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Practical No.:1

Demonstrate basic Git commands

1. Check git version

git --version

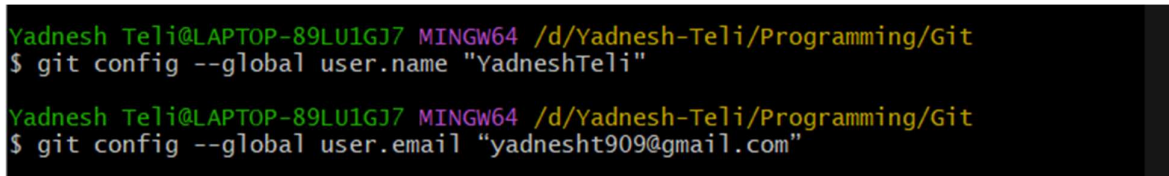
A terminal window titled 'MINGW64:/d/Yadnesh-Teli/Programming/Git' showing the command 'git --version' and its output 'git version 2.45.1.windows.1'.

```
MINGW64:/d/Yadnesh-Teli/Programming/Git
Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git
$ git --version
git version 2.45.1.windows.1
```

2. Configure Git

git config --global user.name "usernewncrd"

git config --global user.email "symca669@gmail.com"

A terminal window showing two 'git config' commands being executed to set the global user name and email.

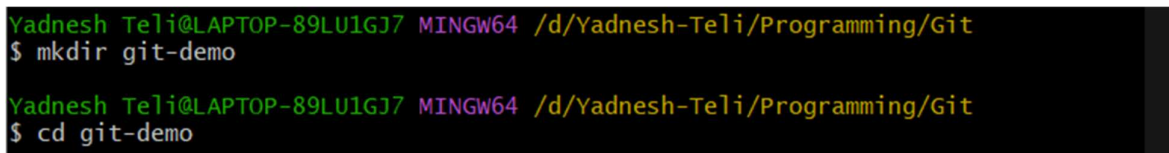
```
Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git
$ git config --global user.name "YadneshTeli"

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git
$ git config --global user.email "yadnesht909@gmail.com"
```

3. Creating a new project folder

mkdir git-demo

cd git-demo

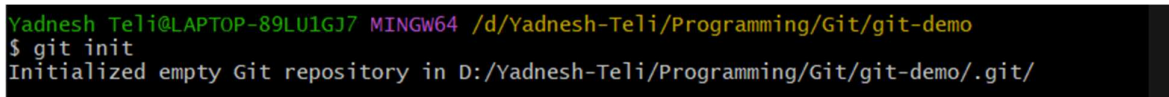
A terminal window showing the 'mkdir git-demo' and 'cd git-demo' commands being executed.

```
Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git
$ mkdir git-demo

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git
$ cd git-demo
```

4. Initialize Git

git init

A terminal window showing the 'git init' command being executed, resulting in an initialized empty Git repository.

```
Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo
$ git init
Initialized empty Git repository in D:/Yadnesh-Teli/Programming/Git/git-demo/.git/
```

5. Git Operations

echo "Hello World">file.txt

git add .

git commit -m "Initial Commit"

```
Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo (master)
$ echo "Hello World">file.txt

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo (master)
$ git add .
warning: in the working copy of 'file.txt', LF will be replaced by CRLF the next time Git touches it

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo (master)
$ git commit -m "Initial Commit"
[master (root-commit) 297fbfa] Initial Commit
1 file changed, 1 insertion(+)
create mode 100644 file.txt
```

6. Checks

git status

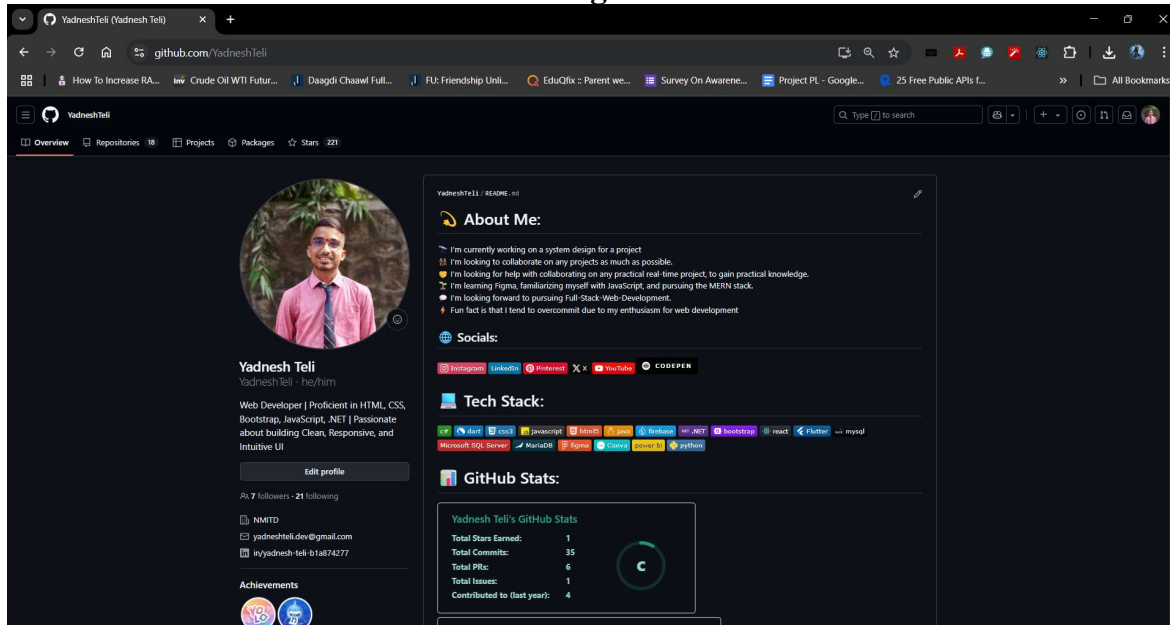
git log

```
Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo (master)
$ git status
On branch master
nothing to commit, working tree clean

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo (master)
$ git log
commit 297fbfaf4227de6ec961865ee37a60af07c16e58 (HEAD -> master)
Author: YadneshTeli <"yadnesht909@gmail.com">
Date: Tue May 20 13:52:32 2025 +0530

    Initial Commit
```

1. Create a GitHub account and log in.



2. Create a repository on GitHub (e.g., git-practice).

The image shows two screenshots of the GitHub web interface. The top screenshot is the 'Create a new repository' page. It has a dark theme. The 'Owner' is 'YadneshTeli' and the 'Repository name' is 'tesst'. A green checkmark indicates 'tesst is available'. The 'Description' field is empty. The 'Public' radio button is selected. Under 'Initialize this repository with:', the 'Add a README file' checkbox is checked. The 'Add .gitignore' section shows a dropdown set to 'None'. The 'Choose a license' section also shows a dropdown set to 'None'. A green 'Create repository' button is at the bottom right.

The bottom screenshot shows the newly created repository 'tesst' on the GitHub page. It includes sections for 'Start coding with Codespaces', 'Add collaborators to this repository', and 'Quick setup — if you've done this kind of thing before'. The quick setup section provides instructions for setting up the repository on a desktop or via the command line.

```
...or create a new repository on the command line

echo "# tesst" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/YadneshTeli/tesst.git
git push -u origin main

...or push an existing repository from the command line

git remote add origin https://github.com/YadneshTeli/tesst.git
git branch -M main
git push -u origin main
```

ProTip! Use the URL for this page when adding GitHub as a remote.

3. Fork any public repository or your own from another account

The image shows two screenshots of the GitHub web interface. The top screenshot displays the 'Create a new fork' page for the repository 'Vineet90x/bookstoreweb'. The page includes a form to create a fork, with fields for 'Owner' (YadneshTeli) and 'Repository name' (bookstoreweb-fork). A green checkmark indicates that 'bookstoreweb-fork' is available. Below the form, there is a checkbox for 'Copy the master branch only' and a note about creating a fork in a personal account. A green 'Create fork' button is at the bottom right.

The bottom screenshot shows the resulting 'bookstoreweb-fork' repository page under the user 'YadneshTeli'. The page is public and shows the repository's history, including a commit titled 'title updated' by 'YadneshTeli' 10 months ago. The repository has 1 branch (master) and 0 tags. The 'About' section on the right indicates that there is no description, website, or topics provided. The 'Releases' and 'Packages' sections also show no published content.

4. Clone the forked repo:

mkdir fork_test && cd fork_test

git clone <https://github.com/YadneshTeli/bookstoreweb-fork.git>

```
Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo (master)
$ mkdir fork_test && cd fork_test

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/fork_test (master)
$ git clone https://github.com/YadneshTeli/bookstoreweb-fork.git
Cloning into 'bookstoreweb-fork'...
remote: Enumerating objects: 1058, done.
remote: Counting objects: 100% (1058/1058), done.
remote: Compressing objects: 100% (829/829), done.
remote: Total 1058 (delta 167), reused 1054 (delta 163), pack-reused 0 (from 0)
Receiving objects: 100% (1058/1058), 32.28 MiB | 564.00 KiB/s, done.
Resolving deltas: 100% (167/167), done.
```

```
Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/fork_test (master)
$ cd bookstoreweb-fork

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/fork_test/bookstoreweb-fork (master)
$
```

5. Create a branch:

git checkout -b feature

```
Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/fork_test/bookstoreweb-fork (master)
$ git checkout -b feature
Switched to a new branch 'feature'

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/fork_test/bookstoreweb-fork (feature)
$
```

6. Make changes, then commit:

echo "Feature added" >> newfile.txt

git add .

git commit -m "Added new feature"

```
Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/fork_test/bookstoreweb-fork (feature)
$ echo "Feature Test" >> newfile.txt

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/fork_test/bookstoreweb-fork (feature)
$ git add newfile.txt
warning: in the working copy of 'newfile.txt', LF will be replaced by CRLF the next time Git touches it

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/fork_test/bookstoreweb-fork (feature)
$ git commit -m "Added new feature"
[feature 7fad3b1] Added new feature
1 file changed, 1 insertion(+)
create mode 100644 newfile.txt
```

7. Merge branch into main:

git checkout master

git merge feature

```
Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/fork_test/bookstoreweb-fork
(feature)
$ git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/fork_test/bookstoreweb-fork
(master)
$ git merge feature
Updating 6486e5c..7fad3b1
Fast-forward
 newfile.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 newfile.txt
```

8. Rebase branch (alternative to merge):

git checkout feature

git rebase master

```
Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/fork_test/bookstoreweb-fork
(master)
$ git checkout feature
Switched to branch 'feature'

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/fork_test/bookstoreweb-fork
(feature)
$ git rebase master
Current branch feature is up to date.
```

9. Push to GitHub:

git push origin feature

git checkout master

git push origin master

```
Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/fork_test/bookstoreweb-fork
(feature)
$ git push origin feature
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 301 bytes | 301.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'feature' on GitHub by visiting:
remote:   https://github.com/YadneshTeli/bookstoreweb-fork/pull/new/feature
remote:
To https://github.com/YadneshTeli/bookstoreweb-fork.git
 * [new branch]      feature -> feature

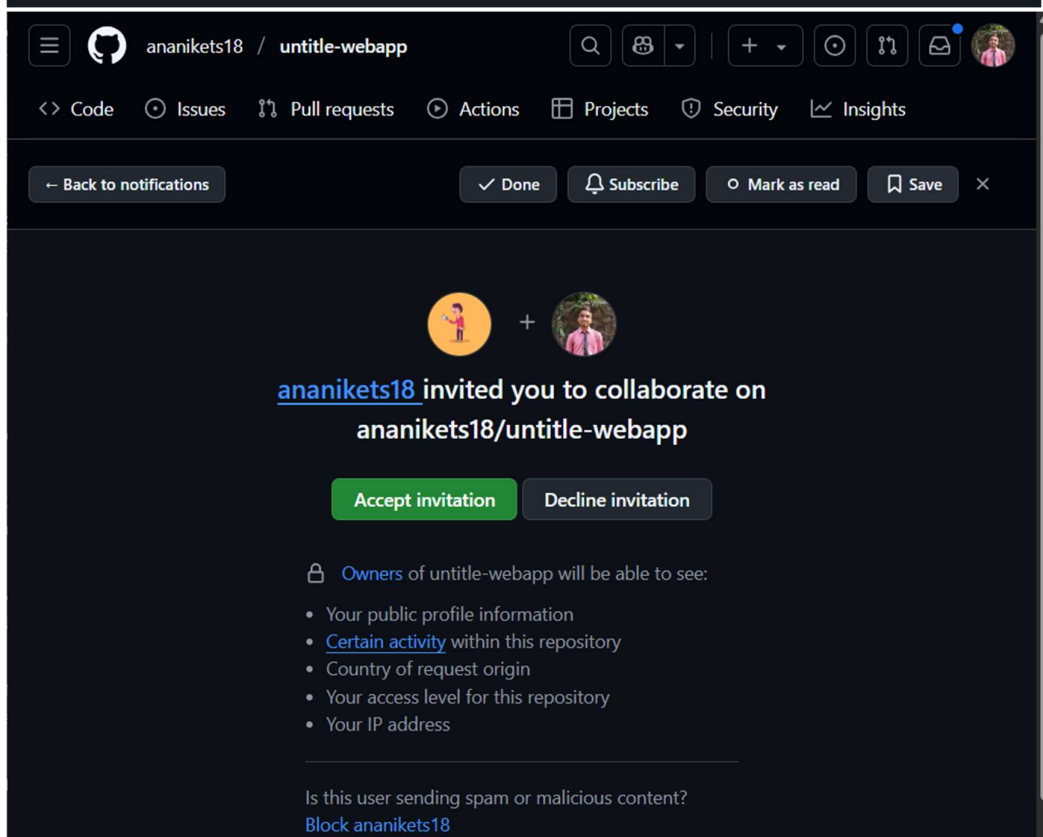
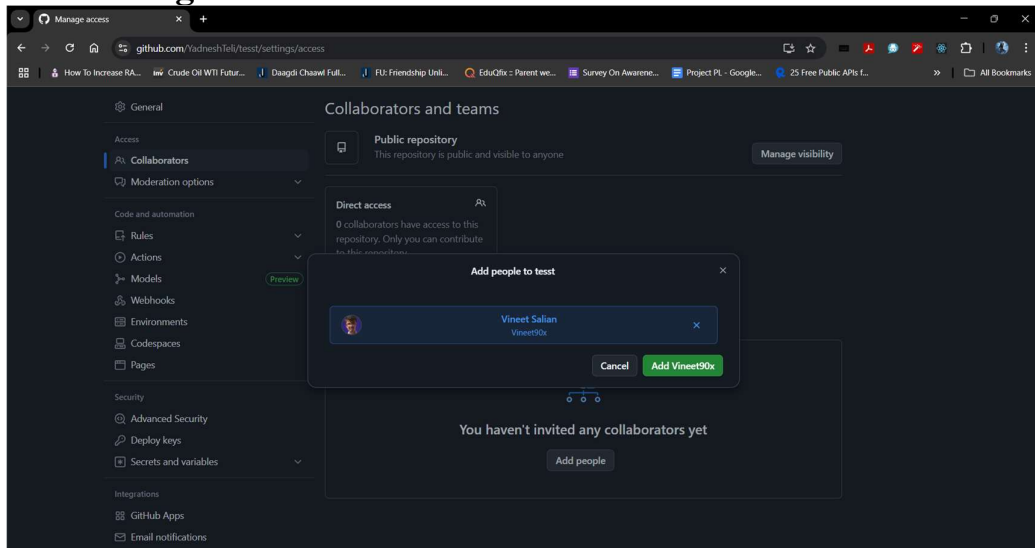
Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/fork_test/bookstoreweb-fork
(feature)
$ git checkout master
Switched to branch 'master'
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/fork_test/bookstoreweb-fork
(master)
$ git push origin master
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/YadneshTeli/bookstoreweb-fork.git
 6486e5c..7fad3b1  master -> master
```

Practical No.:3

Demonstrate Git for Collaboration

1. Using Git for Collaboration



2. Friend clones the repo:

```
git clone https://github.com/YadneshTeli/untitled-webapp.git  
cd team-repo  
git checkout -b bug-fix
```

```

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo (master)
$ mkdir test && cd test

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/test (master)
$ git clone https://github.com/YadneshTeli/untitled-webapp.git
Cloning into 'untitled-webapp'...
remote: Enumerating objects: 1764, done.
remote: Counting objects: 100% (1764/1764), done.
remote: Compressing objects: 100% (1398/1398), done.
remote: Total 1764 (delta 249), reused 1759 (delta 244), pack-reused 0 (from 0)
Receiving objects: 100% (1764/1764), 27.01 MiB | 99.00 KiB/s, done.
Resolving deltas: 100% (249/249), done.
Updating files: 100% (1579/1579), done.

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/test (master)
$ cd untitled-webapp

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/test/untitled-webapp (main)
$ git checkout -b bug-fix
Switched to a new branch 'bug-fix'

```

3. Friend makes changes and pushes:

```

echo "Bug fixed" >> bug.txt
git add .
git commit -m "Fixed a bug"

```

```

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/test/untitled-webapp (bug-fix)
$ echo "Bug fixed" >> bug-test.txt

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/test/untitled-webapp (bug-fix)
$ git add .
warning: in the working copy of 'bug-test.txt', LF will be replaced by CRLF the next time Git touches it

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/test/untitled-webapp (bug-fix)
$ git commit -m "Fixed a bug"
[bug-fix b2f921f] Fixed a bug
1 file changed, 1 insertion(+)
create mode 100644 bug-test.txt

```

4. Push

```
git push origin bug-fix
```

```

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/test/untitled-webapp (bug-fix)
$ git push origin bug-fix
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 302 bytes | 302.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'bug-fix' on GitHub by visiting:
remote:   https://github.com/YadneshTeli/untitled-webapp/pull/new/bug-fix
remote:
To https://github.com/YadneshTeli/untitled-webapp.git
 * [new branch]      bug-fix -> bug-fix

```

```

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/test/untitle-webapp (bug-fix)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/test/untitle-webapp (main)
$ git merge bug-fix
Updating 5ab2a6c..b2f921f
Fast-forward
 bug-test.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 bug-test.txt

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/test/untitle-webapp (main)
$ git push origin bug-fix
Everything up-to-date

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/test/untitle-webapp (main)
$ git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/YadneshTeli/untitle-webapp.git
 5ab2a6c..b2f921f  main -> main

Yadnesh Teli@LAPTOP-89LU1GJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/git-demo/test/untitle-webapp (main)
$ |

```

Comparing changes

Choose two branches to see what's changed or to start a new pull request. If you need to, you can also [compare across forks](#) or [learn more about diff comparisons](#).

base repository: ananikets18/untitle-webapp
base: main
head repository: YadneshTeli/untitle-webapp
compare: bug-fix

✓ Able to merge. These branches can be automatically merged.

Add a title

Reviewers

No reviews

Add a description

Write

Preview

Add your description here...

Assignees

No one—assign yourself

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Use [closing keywords](#) in the description to automatically close issues

☒ Allow edits by maintainers

Create pull request

Helpful resources

ananikets18 / untitled-webapp
Search

Code
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Projects
Wiki
Security
Insights

Fixed a bug #1

YadneshTeli wants to merge 1 commit into ananikets18:main from YadneshTeli:bug-fix

Conversation

Commits

Checks

Files changed

YadneshTeli commented now

No description provided.

Fixed a bug

b2f921f

✓ No conflicts with base branch

Merging can be performed automatically.

Merge pull request

Reviewers

No reviews

Still in progress? [Convert to draft](#)

Assignees

No one—assign yourself

Labels

None yet

Projects

None yet

Milestone

No milestone

Practical No.:4

Demonstrate Collaborating and cloning using Git

1. Clone a public repository:

git clone <https://github.com/YadneshTeli/tesst.git>

```
Yadnesh Teli@LAPTOP-89LUIGJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git
$ git clone https://github.com/YadneshTeli/tesst.git
Cloning into 'tesst'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
```

2. Create a branch:

git checkout -b update-readme

```
Yadnesh Teli@LAPTOP-89LUIGJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git
$ cd tesst

Yadnesh Teli@LAPTOP-89LUIGJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/tesst (main)
$ git checkout -b update-readme
Switched to a new branch 'update-readme'
```

3. Edit and commit changes:

echo "Added a line" >> README.md

git add README.md

git commit -m "Updated README"

```
Yadnesh Teli@LAPTOP-89LUIGJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/tesst (update-readme)
$ echo "Added a line" >> README.md

Yadnesh Teli@LAPTOP-89LUIGJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/tesst (update-readme)
$ git add README.md
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it

Yadnesh Teli@LAPTOP-89LUIGJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/tesst (update-readme)
$ git commit -m "Updated README"
[update-readme 672ab1c] Updated README
1 file changed, 1 insertion(+)
create mode 100644 README.md
```

4. Push and open pull request:

```
Yadnesh Teli@LAPTOP-89LUIGJ7 MINGW64 /d/Yadnesh-Teli/Programming/Git/tesst (update-readme)
$ git push origin update-readme
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 310 bytes | 310.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'update-readme' on GitHub by visiting:
remote:   https://github.com/YadneshTeli/tesst/pull/new/update-readme
remote:
To https://github.com/YadneshTeli/tesst.git
 * [new branch]      update-readme -> update-readme
```

Updated README #1

Open

YadneshTeli wants to merge 1 commit into `main` from `update-readme`

Conversation 0

Commits 1

Checks 0

Files changed 1

YadneshTeli commented now

readme update

Updated README

No conflicts with base branch

Merging can be performed automatically.

Merge pull request

No reviews

Still in progress? [Convert to draft](#)

No one assigned yourself

Labels

None yet

Projects

None yet

YadneshTeli / testt

Type to search

<> Code

Issues

Pull requests 1

Actions

Projects

Wiki

Security

Insights

Settings

Updated README #1

Merged

YadneshTeli merged 1 commit into `main` from `update-readme` [now](#)

Conversation 0

Commits 1

Checks 0

Files changed 1

+1 -0

YadneshTeli commented 1 minute ago

Owner

...

readme update

Updated README

672ab1c

YadneshTeli merged commit 3c81c04 into `main` [now](#)

Revert

Pull request successfully merged and closed

Delete branch

You're all set — the `update-readme` branch can be safely deleted.

Add a comment

Reviewers

No reviews

Still in progress? [Convert to draft](#)

Assignees

No one—[assign yourself](#)

Labels

None yet

Projects

None yet

Milestone

No milestone

Practical No.:5

Using GitLab Web IDE

1. Sign up at <https://gitlab.com>
2. Create a project.
3. Click on Web IDE in your repository.

The screenshot shows the 'Create blank project' page in GitLab. The left sidebar contains navigation links: Your work, Projects, Groups, Issues, Merge requests, To-Do List, Milestones, Snippets, Activity, Import history, Workspaces, Environments, Operations, Security, and Help. The main content area is titled 'Create blank project' and includes a description: 'Create a blank project to store your files, plan your work, and collaborate on code, among other things.' Below this, there are fields for 'Project name' (containing 'test-proj'), 'Project URL' (https://gitlab.com/ronin1321518), and 'Project slug' (test-proj). There is also a 'Project deployment target (optional)' dropdown set to 'Select the deployment target'. Under 'Visibility Level', the 'Private' option is selected, with a note: 'Project access must be granted explicitly to each user. If this project is part of a group, access is granted to members of the group.' Other options are 'Internal' and 'Public'. The 'Project Configuration' section has a checked box for 'Initialize repository with a README' and an unchecked box for 'Enable Static Application Security Testing (SAST)'.

Your work / Projects / New project / Create blank project

Create blank project

Create a blank project to store your files, plan your work, and collaborate on code, among other things.

Project name
test-proj
Must start with a lowercase or uppercase letter, digit, emoji, or underscore. Can also contain dots, pluses, dashes, or spaces.

Project URL
https://gitlab.com/ronin1321518

Project slug
test-proj

Project deployment target (optional)
Select the deployment target

Visibility Level
☒ Private
Project access must be granted explicitly to each user. If this project is part of a group, access is granted to members of the group.
☐ Internal
The project can be accessed by any logged in user except external users.
☐ Public
The project can be accessed without any authentication.

Project Configuration
☒ Initialize repository with a README
Allows you to immediately clone this project's repository. Skip this if you plan to push up an existing repository.
☐ Enable Static Application Security Testing (SAST)

The screenshot shows the project page for 'test-proj' in GitLab. The left sidebar contains navigation links: Project, Learn GitLab (23%), Pinned, Issues (0), Merge requests (0), Manage, Plan, Code, Build, Secure, Deploy, Operate, Monitor, and Help. The main content area shows a notification: 'You can't push or pull repositories using SSH until you add an SSH key to your profile.' Below this, a message states: 'Project 'test-proj' was successfully created.' The project name 'test-proj' is displayed with a 'Free' label. The 'main' branch is selected. A table shows the 'Initial commit' by 'Yadnesh Teli' just now. Below the table, there is a 'README.md' file. The 'Project information' sidebar on the right shows: 1 Commit, 1 Branch, 0 Tags, 4 KIB Project Storage, and links to add LICENSE, CHANGELOG, CONTRIBUTING, enable Auto DevOps, and add a Kubernetes cluster.

RoNIN / test-proj

You can't push or pull repositories using SSH until you add an SSH key to your profile.
[Add SSH key](#) [Don't show again](#)

Project 'test-proj' was successfully created.

test-proj

main test-proj

Initial commit
Yadnesh Teli authored just now
fb3c3a9f History

Name	Last commit	Last update
README.md	Initial commit	just now

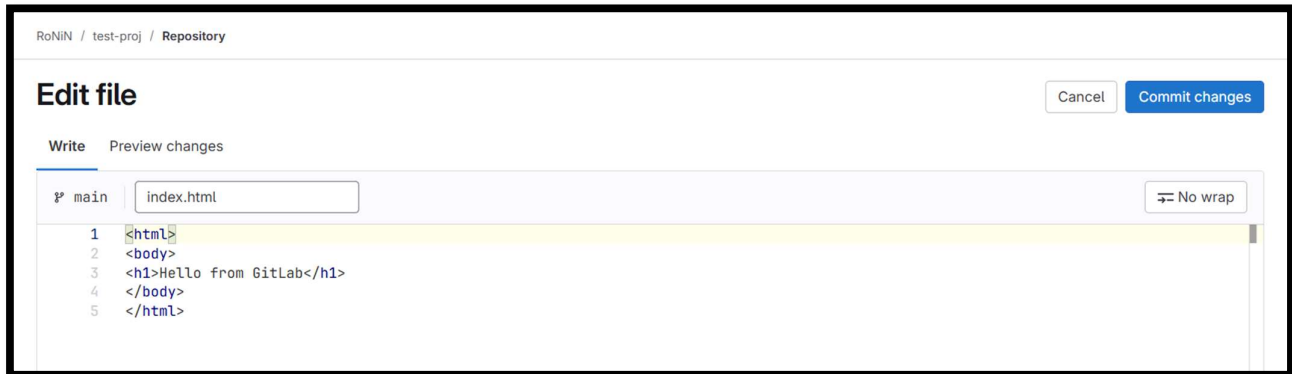
test-proj

Getting started

Project information
1 Commit
1 Branch
0 Tags
4 KIB Project Storage
README
+ Add LICENSE
+ Add CHANGELOG
+ Add CONTRIBUTING
+ Enable Auto DevOps
+ Add Kubernetes cluster

4. Create a file (index.html):

```
<html>
  <body>
    <h1>Hello from GitLab</h1>
  </body>
</html>
```



5. Click Commit and push changes:



Practical No.:6

Demonstrate CI/CD Workflow in GitLab using .py, .bash, .java file

Bash

RoNIN / Test69 / General Settings

Q Search page

▼ **Naming, description, topics**
Update your project name, description, avatar, and topics. [Learn more about projects.](#)

Project name
Test69
Must start with letter, digit, emoji, or underscore. Can also contain periods, dashes, spaces, and parentheses.

Project ID
69347481

Project avatar
 Choose file... No file chosen.
The ideal image size is 192 x 192 pixels. The maximum file size allowed is 200 KiB.

Project description (optional)

Project topics
Search for topic
Topics are publicly visible even on private projects. Do not include sensitive information in topic names. [Learn more.](#)

```
echo "This is from my bash script"
touch myFile.txt
echo "Helloo 69" > myFile.txt
echo "Testt"
```

RoNIN / Test69

⚠ You can't push or pull repositories using SSH until you add an SSH key to your profile. [Add SSH key](#) [Don't show again](#)

main test69 / basic.sh

basic.sh Find file Blame Edit

Add new file
Yadnesh Teli authored 3 weeks ago 79c480f8 History

basic.sh 97 B

```
1 echo "This is from my bash script"
2 touch myFile.txt
3 echo "Helloo 69" > myFile.txt
4 echo "Testt"
```

```
stages:
  - build
bash_execute:
  stage: build
script:
  - bash ./basic.sh
```

⚠ You can't push or pull repositories using SSH until you add an SSH key to your profile.

[Add SSH key](#)[Don't show again](#)

main

test69 / .gitlab-ci.yml

.gitlab-ci.yml

[Find file](#)[Blame](#)[Edit](#)

Add new file

Yadnesh Teli authored 3 weeks ago



7af4733d

[History](#)

.gitlab-ci.yml 93 B



```
1 stages:
2   - build
3 bash_execute:
4   stage: build
5   script:
6     - bash ./basic.sh
```

All 1

[Finished](#)[Branches](#)[Tags](#)[View analytics](#)[Clear runner caches](#)[New pipeline](#)[Show Pipeline ID](#)

Status

Pipeline

Created by

Stages

Actions

Passed

 00:00:29
 3 weeks ago

Add new file

#1789264868

main

7af4733d



latest

branch

PYTHON

RoNIN / nagoba / General Settings

Q Search page

▼ Naming, description, topics

Update your project name, description, avatar, and topics. [Learn more about projects.](#)

Project name

nagoba

Project ID

69348019

Must start with letter, digit, emoji, or underscore. Can also contain periods, dashes, spaces, and parentheses.

Project avatar

N

Choose file... No file chosen.

The ideal image size is 192 x 192 pixels. The maximum file size allowed is 200 KiB.

Project description (optional)

Project topics

Search for topic

Topics are publicly visible even on private projects. Do not include sensitive information in topic names. [Learn more.](#)

script.py

```
print("HELLO FROM NAGOBAB")
```

RoNIN / nagoba

You can't push or pull repositories using SSH until you add an SSH key to your profile.

Add SSH key

Don't show again

main ▼

nagoba / script.py

script.py

Find file

Blame

Edit ▼

Add new file

Yadnesh Teli authored 3 weeks ago

cdf8db2d

History

script.py 26 B

1

print("HELLO FROM NAGOBAB")

.gitlab-ci.yml

```
stages:
  - test

python_script:
  stage: test
  image: python:3.10
  script:
    - python script.py
```

⚠ You can't push or pull repositories using SSH until you add an SSH key to your profile.

[Add SSH key](#)[Don't show again](#)

main nagoba / .gitlab-ci.yml

.gitlab-ci.yml

[Find file](#)[Blame](#)[Edit](#)[Edit .gitlab-ci.yml](#)

Yadnesh Teli authored 3 weeks ago



702d99a8

[History](#)

.gitlab-ci.yml 119 B



```
1 stages:
2   - test
3
4 python_script:
5   stage: test
6   image: python:3.18
7   script:
8     - python script.py
```

[All](#)

3

[Finished](#)[Branches](#)[Tags](#)[View analytics](#)[Clear runner caches](#)[New pipeline](#)[Show Pipeline ID](#)

Status	Pipeline	Created by	Stages	Actions
Passed 00:00:29 3 weeks ago	Edit .gitlab-ci.yml #1789275415 main 702d99a8 latest branch			Download
Failed 00:00:29 3 weeks ago	Update .gitlab-ci.yml file #1789274625 main a3dce7e6 branch			Refresh Download
Failed 3 weeks ago	Add new file #1789273433 main dacfc8c6 error branch			Download

JAVA

RoNIN / Javaaaaa / General Settings

▼ Naming, description, topics

Update your project name, description, avatar, and topics. [Learn more about projects.](#)

Project name

Javaaaaa

Must start with letter, digit, emoji, or underscore. Can also contain periods, dashes, spaces, and parentheses.

Project ID

69348586

Project avatar

J

Choose file...

No file chosen.

The ideal image size is 192 x 192 pixels. The maximum file size allowed is 200 KiB.

Project description (optional)

Project topics

Search for topic

Topics are publicly visible even on private projects. Do not include sensitive information in topic names. [Learn more.](#)

JAvaaa.java

```
class JAvaaa{
    public static void main(String a[]){
        System.out.println("Hello World!");
        System.out.println("Agent 47 arrived in Lahore!");
    }
}
```

RoNIN / Javaaaaa

⚠ You can't push or pull repositories using SSH until you add an SSH key to your profile.

Add SSH key

Don't show again

main ▼

javaaaaa / JAvaaa.java

JAvaaa.java

Find file

Blame

Edit ▼

Edit JAvaaa.java

Yadnesh Teli authored 3 weeks ago

0c6836e2

History

JAvaaa.java 172 B

```
1 class JAvaaa{
2     public static void main(String a[]){
3         System.out.println("Hello World!");
4         System.out.println("Agent 47 arrived in Lahore!");
5     }
6 }
7
```

.gitlab-ci.yml

stages:

- build
- test

before_script:

- apt-get update && apt-get install -y openjdk-17-jdk

build:

stage: build

script:

- javac JAvaaa.java
- ls -ls

artifacts:

paths:

- JAvaaa.class

only:

- main

test:

stage: test

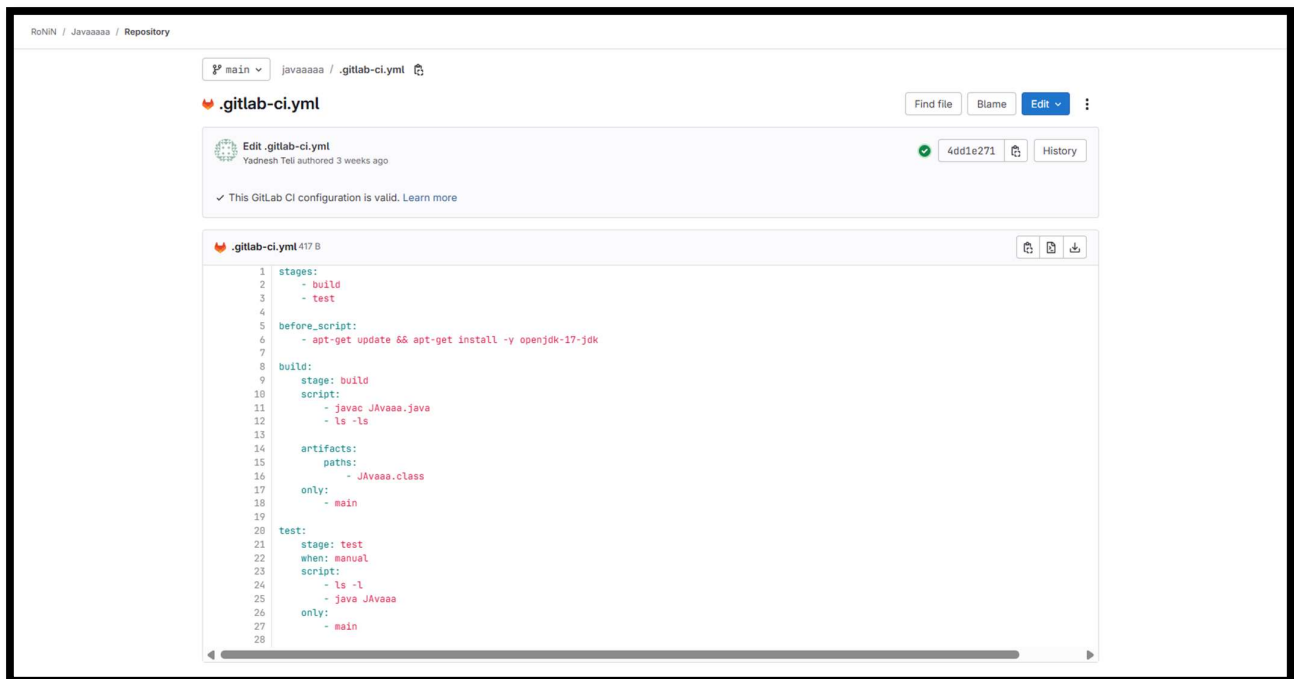
when: manual

script:

- ls -l
- java JAvaaa

only:

- main



All 5 Finished Branches Tags

[View analytics](#)

[Clear runner caches](#)

[New pipeline](#)

Filter pipelines



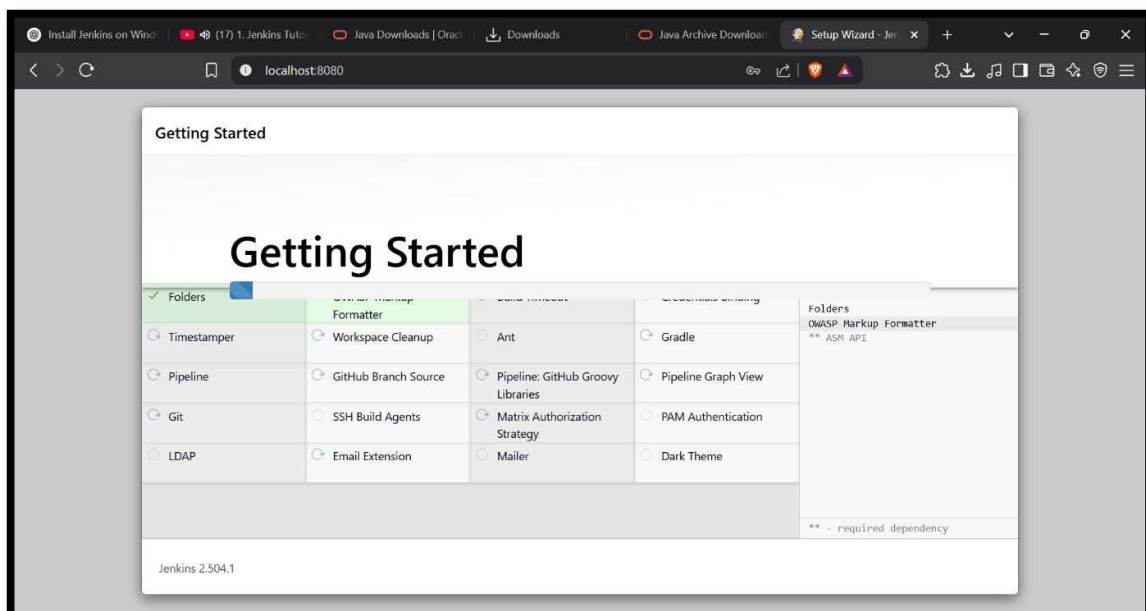
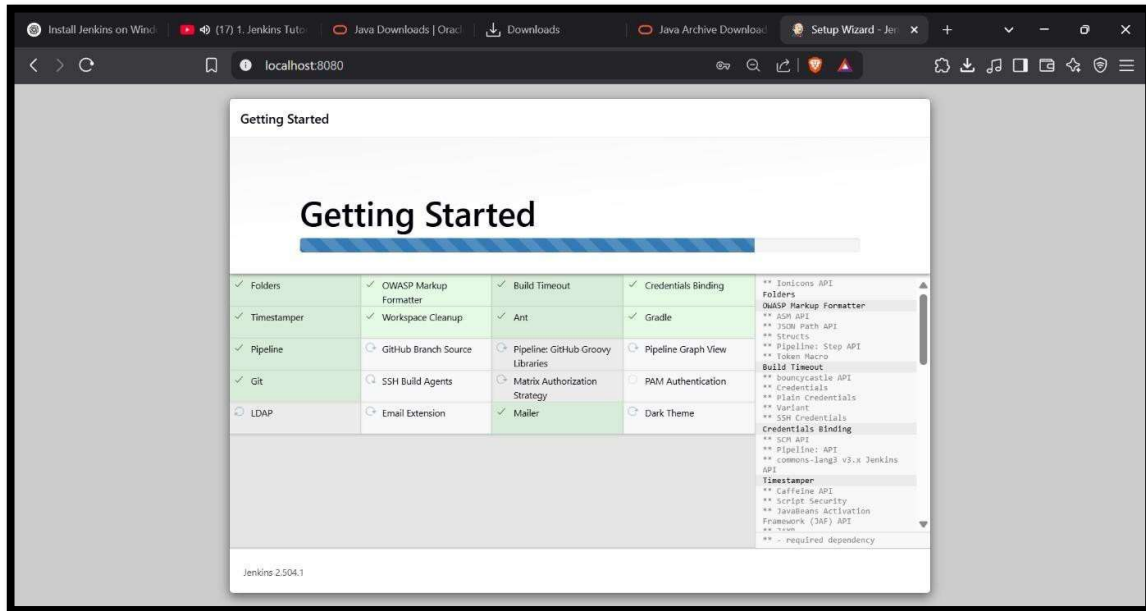
Show Pipeline ID

Status	Pipeline	Created by	Stages	Actions
Passed 00:01:54 3 weeks ago	Edit .gitlab-ci.yml #1789305809 main 4dd1e271 latest branch		✓ ✓	▶ ⬇
Failed 00:00:55 3 weeks ago	Edit JAvaaa.java #1789305511 main 8c6836e2 branch		✗ »	▶ ⬇ ↺ ⬇
Failed 00:00:55 3 weeks ago	Edit .gitlab-ci.yml #1789304615 main 757a71a3 branch		✗ »	▶ ⬇ ↺ ⬇
Failed 3 weeks ago	Add new file #1789302927 main a3568aa9 error branch			⬇
Failed 3 weeks ago	Add new file #1789301822 main 55d91db6 error branch			⬇

Practical No.:7

Demonstrate setting Jenkins CI/CD pipeline.

1. Install Jenkins (visit <https://www.jenkins.io>)
2. Run Jenkins: <http://localhost:8080>



3. Create a new Pipeline project: CI-CD-Demo

Enter an item name

CI-CD-Demo

Select an item type

 **Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

 **Pipeline**
Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.

 **Multi-configuration project**
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

 **Folder**
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

 **Multibranch Pipeline**
Creates a set of Pipeline projects according to detected branches in one SCM repository.

 **Organization Folder**
Creates a set of multibranch project subfolders by scanning for repositories.

OK

Dashboard > CI-CD-Demo > Configuration

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git

Repositories

Repository URL

https://github.com/YadneshTel/test.git

Specify the URL or path of the git repository. This uses the same syntax as your `git clone` command. (from Git plugin)

Credentials

YadneshTel/*****

+ Add

Advanced

Add Repository

Branches to build

Branch Specifier (blank for 'any')

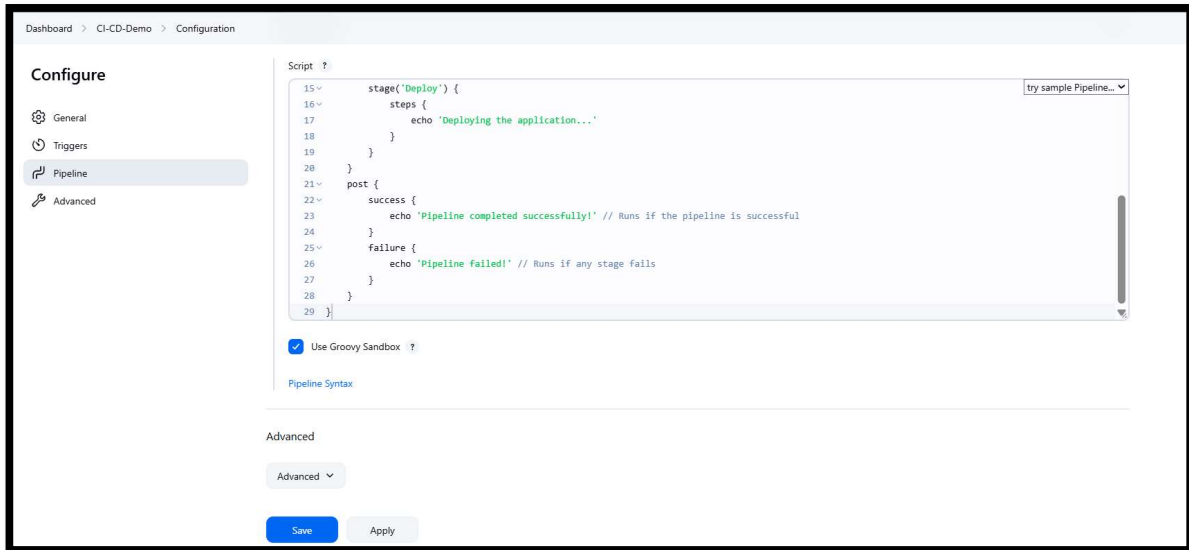
*/main

Save Apply

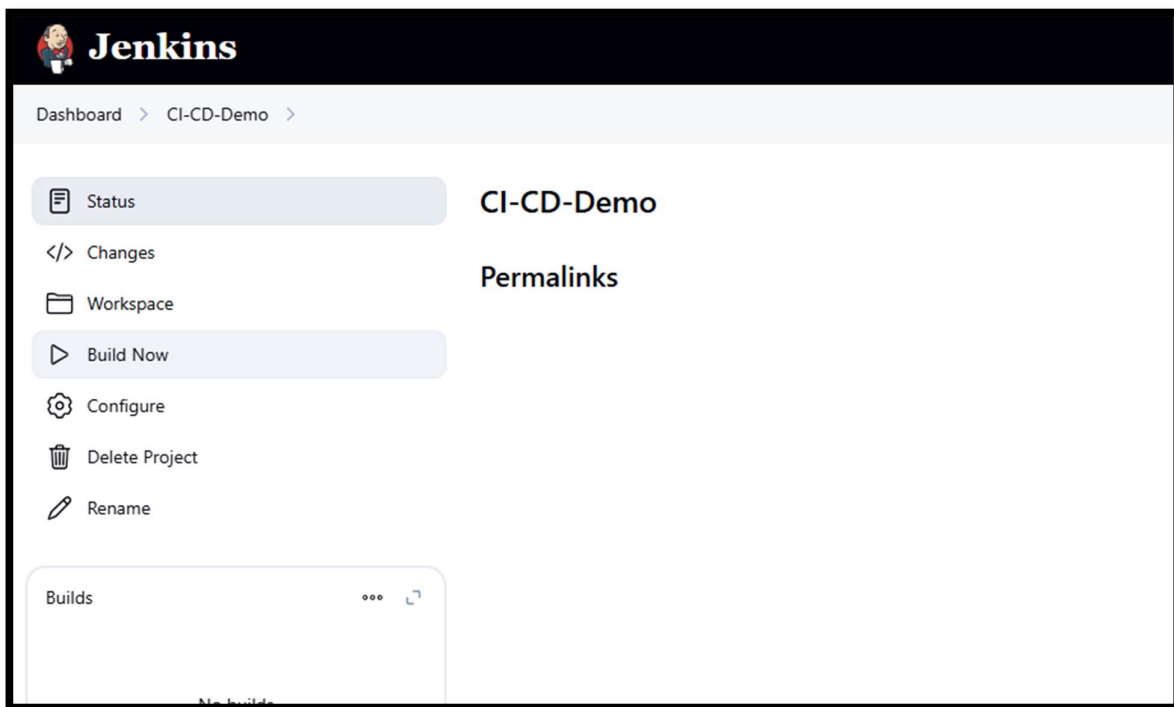
4. Add Pipeline Script > Script:

```
pipeline {
  agent any // Defines where the pipeline runs

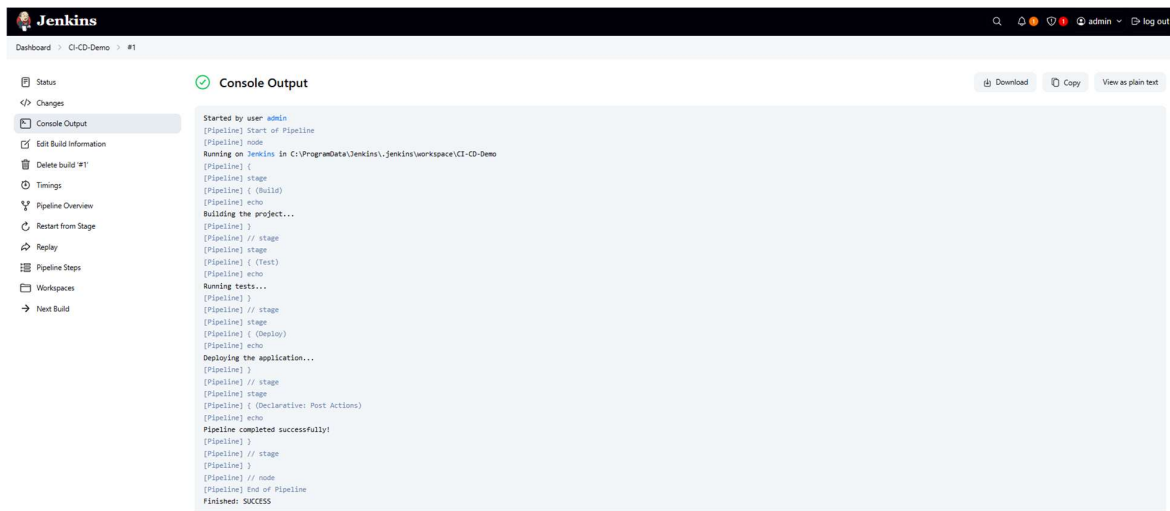
  stages {
    stage('Build') { // Defines a step in the pipeline
      steps {
        echo 'Building the project..' // Print message to console
      }
    }
    stage('Test') {
      steps {
        echo 'Running tests...'
      }
    }
    stage('Deploy') {
      steps {
        echo 'Deploying the application...'
      }
    }
  }
  post {
    success {
      echo 'Pipeline completed successfully!' // Runs if the pipeline is successful
    }
    failure {
      echo 'Pipeline failed!' // Runs if any stage fails
    }
  }
}
```



5. Save and click Build Now.



6. Check output in Console Output.



The screenshot displays the Jenkins web interface. The top navigation bar includes the Jenkins logo, a search icon, and user information (admin) with a log out link. The breadcrumb trail shows the path: Dashboard > CI-CD-Demo > #1. On the left sidebar, the 'Console Output' tab is selected, with other options like Status, Changes, Edit Build Information, Delete build #1, Timings, Pipeline Overview, Restart from Stage, Replay, Pipeline Steps, Workspaces, and Next Build. The main content area, titled 'Console Output' with a green checkmark icon, shows the following log output:

```
Started by user admin
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\workspace\CI-CD-Demo
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Building the project...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Running tests...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploying the application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Pipeline completed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

At the top right of the console output area, there are three buttons: 'Download', 'Copy', and 'View as plain text'.

Practical No.:8

Demonstrate Setting up of a CI/CD pipeline to build and deploy a web application to a local HTTP server

Create a new Dynamic web project (Eclipse IDE for enterprise java and web developers)

Index.jsp :

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8"%>
<html>
<head>
    <title>CookiesDemo</title>
</head>
<body>
    <h2>CookiesDemo - </h2>
    <form action="CookiesDemo.jsp" method="get">
        Name - <input type="text" name="usernm">
        <input type="submit" value="Submit Query">
    </form>
</body>
</html>
```

CookiesDemo.jsp:

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<%@ page import="jakarta.servlet.http.Cookie" %>
s<%@ page import="java.io.*" %>
<html>
<head>
    <title>Session Management using Cookies</title>
</head>
<body>
    <h2>Session Management using Cookies</h2>
    <%
        String username = request.getParameter("usernm");
        Cookie[] cookies = request.getCookies();
        int visitCount = 0;
        boolean userExists = false;

        if (cookies != null) {
            for (Cookie cookie : cookies) {
                if (cookie.getName().equals("visitCount")) {
                    visitCount = Integer.parseInt(cookie.getValue());
                }
                if (cookie.getName().equals("username")) {
                    userExists = true;
                }
            }
        }
        visitCount++;
        Cookie visitCookie = new Cookie("visitCount", String.valueOf(visitCount));
        visitCookie.setMaxAge(60 * 60 * 24);
        response.addCookie(visitCookie);

        if (!userExists && username != null) {
            Cookie userCookie = new Cookie("username", username);
```

```

        userCookie.setMaxAge(60 * 60 * 24);
        response.addCookie(userCookie);
    }
    %>
    <p>Hello <%= username != null ? username : "Guest" %> You have hit the page <%= visitCount %>
time(s)</p>
    <a href="CookiesDemo.jsp?usern=<%= username %>">Hit Again</a>
</body>
</html>

```

Dashboard > All > New Item

Enter an item name

webPipeline

Select an item type

-  **Freestyle project**
Classic, general-purpose job type that checks out from up steps like archiving artifacts and sending email notification
-  **Maven project**
Build a maven project. Jenkins takes advantage of your PO
-  **Pipeline**
Orchestrates long-running activities that can span multiple workflows) and/or organizing complex activities that do n
-  **Multi-configuration project**
Suitable for projects that need a large number of differen

OK

Dashboard > CI-CD-Demo > Configuration

Connect and manage your code repository to automatically pull the latest code for your builds.

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

☐ None

☒ Git

Repositories ?

Repository URL ?

https://github.com/YadneshTeli/test.git

Specify the URL or path of the git repository. This uses the same syntax as your git clone command. (from Git plugin)

Credentials ?

YadneshTeli/*****

+ Add

Advanced ▾

Add Repository

Branches to build ?

Branch Specifier (blank for 'any') ?

*/main

Save Apply

Add Pipeline Script > Script:

```
pipeline {
    agent any

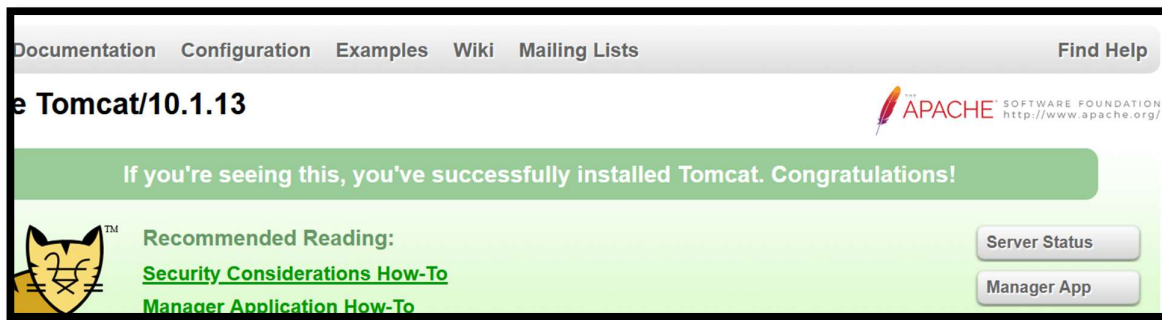
    stages {
        stage('Checkout Code') {
            steps {
                script {
                    git branch: 'master', url: 'https://github.com/YadneshTeli/DevopsJenkins'
                }
            }
        }

        stage('Verify Files') {
            steps {
                bat 'dir /S /B'
            }
        }

        stage('Deploy') {
            steps {
                script {
                    def srcPath = "CookiesDemo/src/main/webapp"
                    def destPath = "C:\\Program Files\\Apache Software Foundation\\Tomcat
11.0\\webapps\\Index"

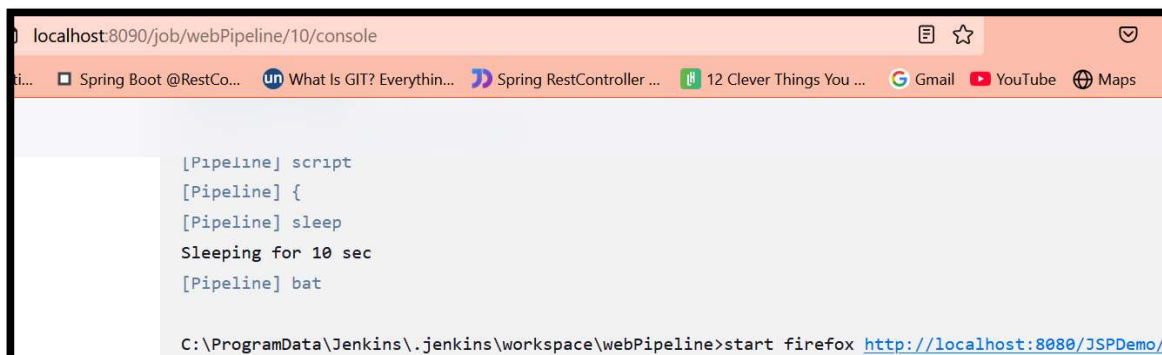
                    if (fileExists(srcPath)) {
                        bat "xcopy /E /I \\\"${srcPath}\\\" \\\"${destPath}\\\""
                    } else {
                        error "Source directory ${srcPath} does not exist!"
                    }
                }
            }
        }
    }
}
```

- Open Manager app from the Tomcat panel by entering username and password –

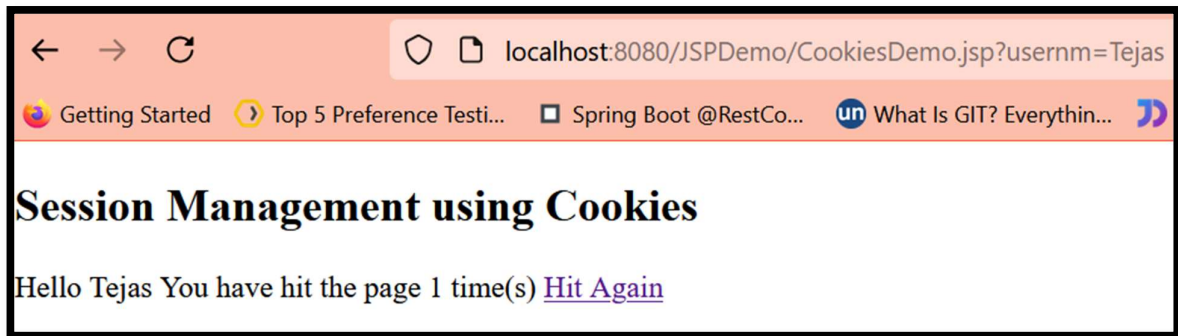


Applications		
Path	Version	Display
/	None specified	Welcome to Tomcat
/JSPDemo	None specified	JSP_Session
/docs	None specified	Tomcat Documentation
/examples	None specified	Servlet and JSP Examples
/host-manager	None specified	Tomcat Host Manager App
/manager	None specified	Tomcat Manager Application

- Click on the link present in the Jenkin's Console Output –



CookiesDemo -
 Name -



Practical No.:9

Demonstrate basic Docker commands

1. Check Docker

version docker --version

```
C:\Users\hp>docker --version
Docker version 28.1.1, build 4eba377
```

2. Pull a Docker image from Docker Hub

docker pull nginx

```
C:\Users\hp>docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
254e724d7786: Pull complete
d38f2ef2d6f2: Pull complete
40a6e9f4e456: Pull complete
913115292750: Pull complete
3e544d53ce49: Pull complete
4f21ed9ac0c0: Pull complete
d3dc5ec71e9d: Pull complete
Digest: sha256:c15da6c91de8d2f436196f3a768483ad32c258ed4e1beb3d367a27ed67253e66
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
```

3. List all Docker images

docker images

```
C:\Users\hp>docker images
REPOSITORY    TAG       IMAGE ID      CREATED        SIZE
nginx         latest    c15da6c91de8  4 weeks ago   279MB
```

4. Run a container from an image

docker run -d -p 8080:80 --name mynginx nginx

This will run the Nginx container and map port 80 (inside the container) to port 8080 (on your host).

```
C:\Users\hp>docker run -d -p 8080:80 --name mynginx nginx
be119c94991613d02f223222f9c1bd038694ab48648d3c1dc38ab8a076e77697
```

5. List all running containers

docker ps

```
C:\Users\hp>docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                    NAMES
be119c949916   nginx     "/docker-entrypoint..." 54 seconds ago Up 54 seconds  0.0.0.0:8080->80/tcp     mynginx
```

6. Copy content from host to container

```
docker cp index.html mynginx:/usr/share/nginx/html/
```

Replace index.html with your actual file. This copies a file into the running container.

```
C:\Users\hp>docker cp index.html mynginx:/usr/share/nginx/html/  
CreateFile C:\Users\hp\index.html: The system cannot find the file specified.
```

7. Create and use Docker volume for persistent content

```
docker volume create mydata
```

```
docker run -d -p 8081:80 --name nginx_vol -v mydata:/usr/share/nginx/html nginx
```

Now any data added to the /usr/share/nginx/html inside the container will persist even if the container is removed.

```
C:\Users\hp>docker run -d -p 8081:80 --name nginx_vol -v mydata:/usr/share/nginx/html nginx  
c8aeab6feb9b69c022e844c4e7e0998bfa40089e9c3d74f64655d4d8c9f99041
```

```
C:\Users\hp>docker volume create mydata  
mydata
```

8. List Docker volumes

```
docker volume ls
```

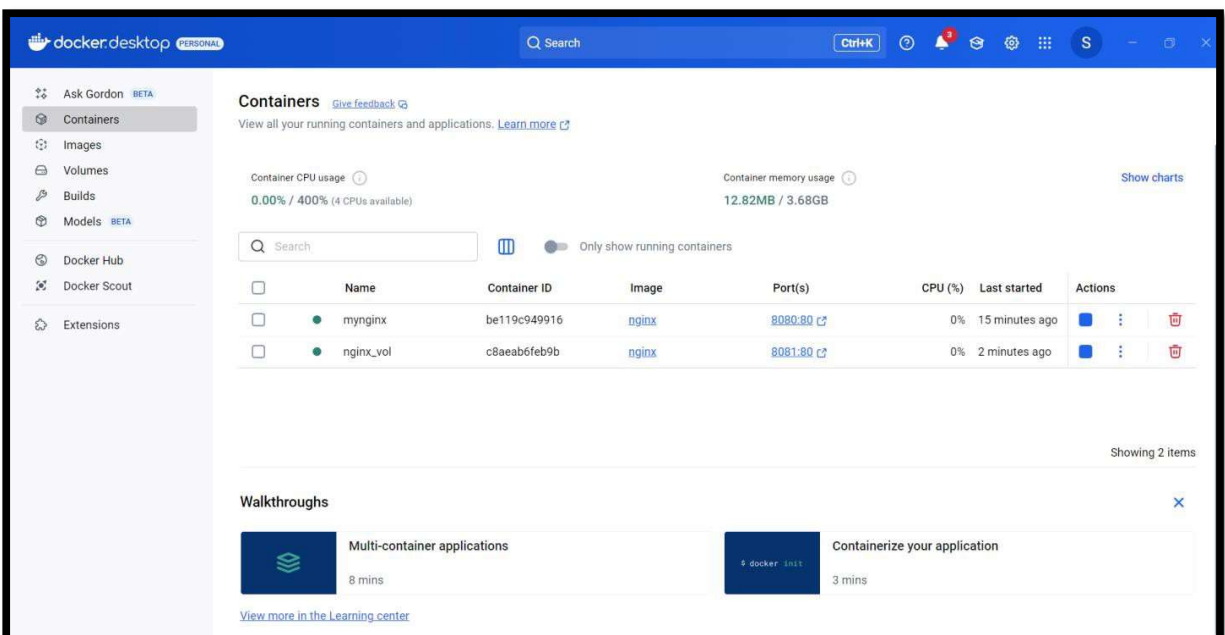
```
C:\Users\hp>docker volume ls  
DRIVER      VOLUME NAME  
local       mydata
```

9. Remove a container

```
docker rm -f mynginx
```

10. Remove an image

```
docker rmi nginx
```



Practical No.:10

Develop a simple containerized application using Docker

Develop a Simple Containerized Application using Docker

1. Index.html

```
index.html > html
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Document</title>
7  </head>
8  <body>
9      <h1>Hello From Docker Container</h1>
10     <h1>Hello from User</h1>
11 </body>
12 </html>
```

2. Dockerfile:-

```
Dockerfile > FROM
1  FROM nginx:latest
2  COPY index.html /usr/share/nginx/html/index.html
```

3. docker build -t my-docker-webapp.

```
PS C:\Users\myapp> code Dockerfile
PS C:\Users\myapp> docker build -t my-docker-webapp .
[+] Building 2.6s (8/8) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 104B
=> [internal] load metadata for docker.io/library/nginx:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 320B
=> [1/2] FROM docker.io/library/nginx:latest@sha256:c15da6c91de8d2f436196f3a768483ad32c258ed4e1beb3d367a27ed67253e66
=> => resolve docker.io/library/nginx:latest@sha256:c15da6c91de8d2f436196f3a768483ad32c258ed4e1beb3d367a27ed67253e66
=> [auth] library/nginx:pull token for registry-1.docker.io
=> [2/2] COPY index.html /usr/share/nginx/html/index.html
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:0931fdbc45611ccb959befd0840ebe19bdae276a8549fe234f5e5a371c4be98
=> => exporting config sha256:af297af0407d6db437d84ba0b2862f7884945673ff9f3684cbebf6a3d81a67b
=> => exporting attestation manifest sha256:42bd5183c690afd8cb7a835d3dbeefc32ce51f47f1e500f223ea7011c57852a5
=> => exporting manifest list sha256:e52dfa33cc70e51ba0dbe044257e359b09f6e579db77e7061a13610f776cddc1
=> naming to docker.io/library/my-docker-webapp:latest
=> => unpacking to docker.io/library/my-docker-webapp:latest
```

3. `docker run -d -p 8080:80 --name webapp-container my-docker-webapp`

```
PS C:\Users\myapp> docker run -d -p 8080:80 --name webapp-container my-docker-webapp
6eef9ba52dcde9c4d40a54043711f3b21db18984ad7aa017f60311481b678660
```

4. `docker ps`

```
PS C:\Users\myapp> docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                    NAMES
c8aeb6feb9b   nginx    "/docker-entrypoint..." 56 minutes ago Up 56 minutes  0.0.0.0:8081->80/tcp    nginx_vol
be119c949916   nginx    "/docker-entrypoint..." About an hour ago Up About an hour  0.0.0.0:8080->80/tcp    mynginx
```

5. `docker stop webapp-container`

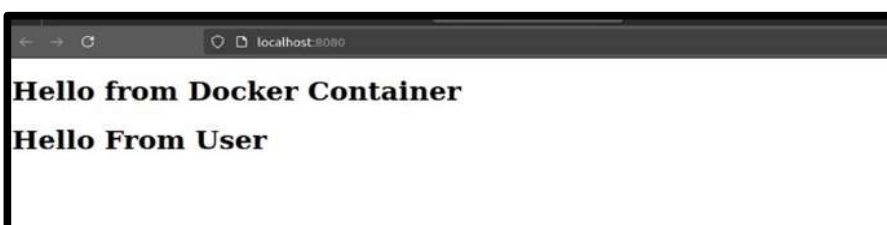
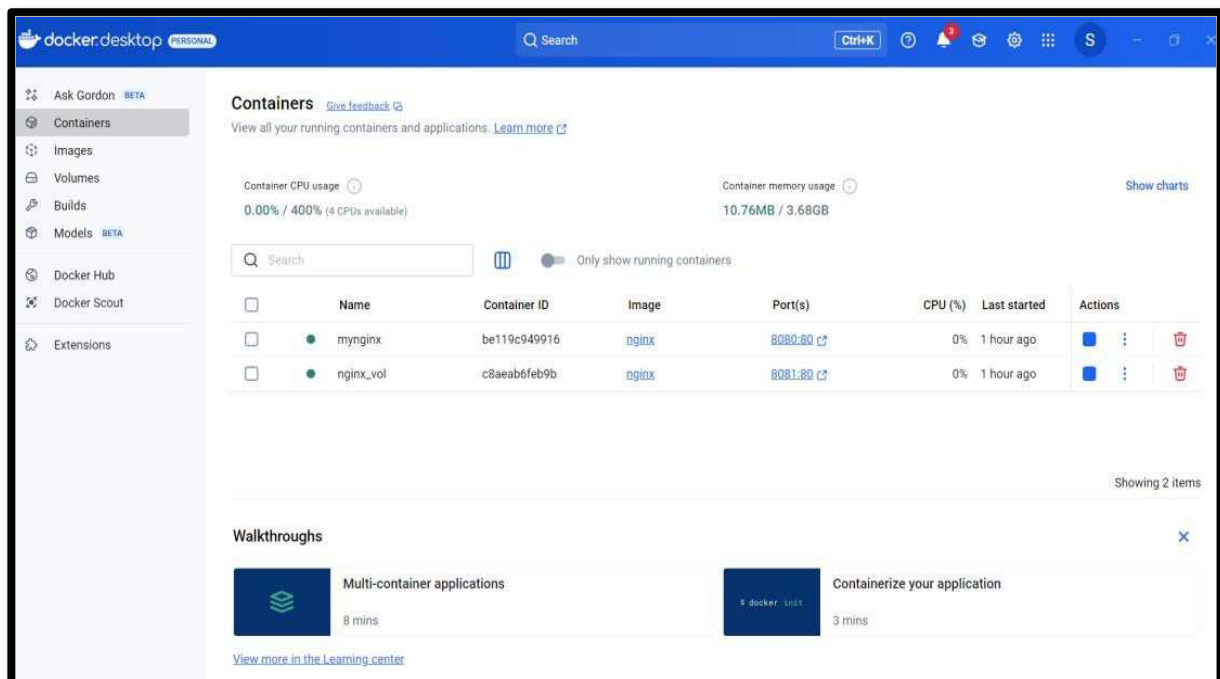
```
PS C:\Users\myapp> docker stop webapp-container
webapp-container
```

6. `docker rm webapp-container`

```
PS C:\Users\myapp> docker rmi my-docker-webapp
Untagged: my-docker-webapp:latest
```

7. `docker rmi my-docker-webapp`

```
PS C:\Users\myapp> docker rmi my-docker-webapp
Untagged: my-docker-webapp:latest
Deleted: sha256:e52dfa33cc70e51ba0dbe044257e359b09f6e579db77e7061a13610f776cddc1
```



Demonstrate add-on ansible commands

Step 1: Update your VM

```
ubuntu@ubuntu:~$ sudo apt update && sudo apt upgrade
[sudo] password for ubuntu:
Hit:1 https://brave-browser-apt-release.s3.brave.com stable InRelease
Ign:2 https://pkg.jenkins.io/debian-stable binary InRelease
Hit:3 https://pkg.jenkins.io/debian-stable binary Release
Hit:4 https://packages.microsoft.com/repos/code stable InRelease
Hit:5 http://security.ubuntu.com/ubuntu focal-security InRelease
Hit:7 http://ppa.launchpad.net/rock-core/at4/ubuntu focal InRelease
Hit:8 http://in.archive.ubuntu.com/ubuntu focal InRelease
Hit:9 http://ppa.launchpad.net/wreshark-dev/stable/ubuntu focal InRelease
Hit:10 http://in.archive.ubuntu.com/ubuntu focal-updates InRelease
Hit:11 http://in.archive.ubuntu.com/ubuntu focal-backports InRelease
Reading package lists... Done
Building dependency tree
Reading state information... Done
67 packages can be upgraded. Run 'apt list --upgradeable' to see them.
Skiping acquire of configured file '/main/binary-i386/Packages' as repository 'https://brave-browser-apt-release.s3.brave.com stable InRelease' doesn't support architecture 'i386'
Reading package lists... Done
Building dependency tree
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
chromium libmpeg-xcsostramer1.0-waspi libgstreser-pubsub-bad1.0-0 libqt5concurrent5 libqt5opengl5-dev libqt5sql5 libqt5sql5-sqlite libqt5test5 libvulkan-dev libwreshark13
libxrender0 libxext0 libxft-dev qt5-gtk-qml qt5-gui qt5-image-plugin qtbase5-dev qtbase5-dev-tools x11proto-next-dev
Use 'dpkg --get-selections' to remove them.
Get new security updates through Ubuntu Pro with 'esm-apps' enabled:
libc-bin libc-vplugin-video-output libavformat58 python2.7-dev libavfilter7
libbdm4 vlc-plugin-samba ipython3 libswresample3 vlc-plugin-qtl libzmq5
python2.7-minimal vlc-plugin-skins vlc-plugin-visualization python-l10n
libgraphviz5 libpython2.7 python2.7 vlc-plugin-notify libvlc5 python3-10n
libpython2.7-dev libqy-plugins-qts libpostproc5 libabcm-dev libvcodec9
libvlbc-bin libmpx2 libzvbi libiwcore5 libid3t5 libavutil56 vlc-data
libpathplan1 libhavedce58 libswscale5 libgyp2 libstdl2-2.0-0 libysofal
libutl-traceroute vlc-plugin-video-splitter libpython2.7-minimal
libgraphviz-dev libqy-vlc-plugin-base libpython2.7-stdlib traceroute
libzvt-common graphviz
Learn more about Ubuntu Pro at https://ubuntu.com/pro
The following NEW packages will be installed:
linux-headers-5.15.0-139-generic linux-hwe-5.15-headers-5.15.0-139 linux-image-5.15.0-139-generic linux-modules-extra-5.15.0-139-generic
The following packages will be upgraded:
linux-headers-5.15.0-139-generic linux-hwe-5.15-headers-5.15.0-139 linux-image-5.15.0-139-generic linux-modules-extra-5.15.0-139-generic
6 packages can be upgraded. Run 'apt list --upgradeable' to see them.
```

Step 2: Install Ansible

[illegible]

Step 3: Check version

```
ubuntu@ubuntu:~$ ansible --version
ansible 2.9.6
  config file = /etc/ansible/ansible.cfg
  configured module search path = ['/home/ubuntu/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  executable location = /usr/bin/ansible
  python version = 3.8.10 (default, Mar 18 2025, 20:04:55) [GCC 9.4.0]
ubuntu@ubuntu:~$
```

```
ubuntu@ubuntu:~$ nano host.ini
ubuntu@ubuntu:~$
```

1. Ping the remote host

ansible local -i host.ini -m ping

```
ubuntu@ubuntu:~$ ansible local -i host.ini -m ping
[DEPRECATION WARNING]: Distribution Ubuntu 20.04 on host localhost should use /usr/bin/python3, but is using /usr/bin/python for backward compatibility with prior Ansible releases. A future
Ansible release will default to using the discovered platform python for this host. See https://docs.ansible.com/ansible/2.9/reference_appendices/interpreter_discovery.html for more
information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
localhost | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python"
  },
  "changed": false,
  "ping": "pong"
}
ubuntu@ubuntu:~$
```

2. Check uptime

ansible local -i host.ini -a "uptime"

```
ubuntu@ubuntu:~$ ansible local -i host.ini -a "uptime"
[DEPRECATION WARNING]: Distribution Ubuntu 20.04 on host localhost should use /usr/bin/python3, but is using /usr/bin/python for backward compatibility with prior Ansible releases. A future
Ansible release will default to using the discovered platform python for this host. See https://docs.ansible.com/ansible/2.9/reference_appendices/interpreter_discovery.html for more
information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
localhost | CHANGED | rc=0 >>
16:31:16 up 2:49, 1 user, load average: 1.08, 0.98, 0.90
ubuntu@ubuntu:~$
```

3. Install a package

ansible local -i host.ini -m apt -a "name=nginx state=present update_cache=yes" – become

```
ubuntu@ubuntu:~$ ansible local -i host.ini -m apt -a "name=nginx state=present update_cache=yes" --become
[DEPRECATION WARNING]: Distribution Ubuntu 20.04 on host localhost should use /usr/bin/python3, but is using /usr/bin/python for backward compatibility with prior Ansible releases. A future
Ansible release will default to using the discovered platform python for this host. See https://docs.ansible.com/ansible/2.9/reference_appendices/interpreter_discovery.html for more
information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
localhost | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python"
  },
  "cache_update_time": 1747566323,
  "cache_updated": true,
  "changed": true,
  "stderr": "",
  "stderr_lines": [],
  "stdout": "Reading package lists...Building dependency tree...Reading state information...The following packages were automatically installed and are no longer required:\n chromium
m-codecs-ffmpeg-extra gstreamer1.0-vaapi\n libgstreamer-plugins-bad1.0-0 libqt5concurrent5 libqt5opengl5-dev libqt5sql5\n libqt5sql5-sqlite libqt5test5 libvulkan-dev libwfreshark13 libwire
tap10\n libxext-dev qt5-gnome qt5-gnome-bins qtbase5-dev\n qtbase5-dev-tools x11proto-xext-dev\n sudo apt autoremove" to remove them.\n\nThe following additional packages will
be installed:\n libnginx-mod-http-image-filter libnginx-mod-http-xslt-filter\n libnginx-mod-mail libnginx-mod-stream nginx-common nginx-core\n\nSuggested packages:\n fcgiwrap nginx-doc\n\nThe following NEW packages will be installed:\n libnginx-mod-http-image-filter libnginx-mod-http-xslt-filter\n libnginx-mod-mail libnginx-mod-stream nginx-common nginx-core\n\n0 upgraded,
7 newly installed, 0 to remove and 67 not upgraded.\nNeed to get 605 kB of archives.\nAfter this operation, 2141 kB of additional disk space will be used.\n\nGet:1 http://ln.archive.ubuntu.c
om/ubuntu focal-updates/main amd64 nginx-common all 1.18.0-0ubuntu1.7 [37.8 kB]\nGet:2 http://ln.archive.ubuntu.com/ubuntu focal-updates/main amd64 libnginx-mod-http-image-filter amd64 1.18.
0-0ubuntu1.7 [14.8 kB]\nGet:3 http://ln.archive.ubuntu.com/ubuntu focal-updates/main amd64 libnginx-mod-http-xslt-filter amd64 1.18.0-0ubuntu1.7 [13.1 kB]\nGet:4 http://ln.archive.ubuntu.co
m/ubuntu focal-updates/main amd64 libnginx-mod-mail amd64 1.18.0-0ubuntu1.7 [43.0 kB]\nGet:5 http://ln.archive.ubuntu.com/ubuntu focal-updates/main amd64 libnginx-mod-stream amd64 1.18.0-bubu
ntui.7 [67.3 kB]\nGet:6 http://ln.archive.ubuntu.com/ubuntu focal-updates/main amd64 nginx-core amd64 1.18.0-0ubuntu1.7 [425 kB]\nGet:7 http://ln.archive.ubuntu.com/ubuntu focal-updates/main
amd64 nginx 1.18.0-0ubuntu1.7 [3620 B]\nPreconfiguring packages ...nfetched 605 kB in 7s (85.6 kB/s)\n\nSelecting previously unselected package nginx-common.\n\n(Reading database ... \n(
Reading database ... 55%\n(Reading database ... 10%\n(Reading database ... 15%\n(Reading database ... 20%\n(Reading database ... 25%\n(Reading database ... 30%\n(Reading database ... 35%\n(Rea
ding database ... 40%\n(Reading database ... 45%\n(Reading database ... 50%\n(Reading database ... 55%\n(Reading database ... 60%\n(Reading database ... 65%\n(Reading database ... 70%\n(Rea
ding database ... 75%\n(Reading database ... 80%\n(Reading database ... 85%\n(Reading database ... 90%\n(Reading database ... 95%\n(Reading database ... 100%\n(Reading database ... 222461 fi
les and directories currently installed.)\n\nPreparing to unpack .../0-nginx-common-1.18.0-0ubuntu1.7.all.deb ... \n\nUnpacking nginx-common (1.18.0-0ubuntu1.7) ... \n\nSelecting previously un
selected package libnginx-mod-http-image-filter.\n\nPreparing to unpack .../1-libnginx-mod-http-image-filter-1.18.0-0ubuntu1.7.and64.deb ... \n\nUnpacking libnginx-mod-http-xslt-filter (1.18.0-0bu
ntui.7) ... \n\nSelecting previously unselected package libnginx-mod-mail.\n\nPreparing to unpack .../3-libnginx-mod-mail-1.18.0-0ubuntu1.7.and64.deb ... \n\nUnpacking libnginx-mod-stream (1.18.0-0bu
ntui.7) ... \n\nSelecting previously unselected package nginx-core.\n\nPreparing to unpack .../5-nginx-core-1.18.0-0ubuntu1.7.and64.deb ... \n\nUnpacking nginx-core (1.18.0-0ubuntu1.7) ... \n
\nSelecting previously unselected package nginx.\n\nPreparing to unpack .../6-nginx-1.18.0-0ubuntu1.7.all.deb ... \n\nUnpacking nginx (1.18.0-0ubuntu1.7) ... \n\nSetting up nginx-common (1.18.0-0bu
ntui.7) ... \n\nCreated symlink /etc/systemd/system/multi-user.target.wants/nginx.service -> /lib/systemd/system/nginx.service.\n\nSetting up libnginx-mod-mail (1.18.0-0ubuntu1.7) ... \n
\nSetting up libnginx-mod-http-image-filter (1.18.0-0ubuntu1.7) ... \n\nSetting up libnginx-mod-stream (1.18.0-0ubuntu1.7) ... \n\nSetting up nginx-core (1.18.0-0ubuntu1.7) ... \n\nSetting up
nginx (1.18.0-0ubuntu1.7) ... \n\nProcessing triggers for systemd (245.4-4ubuntu1.24) ... \n\nProcessing triggers for man-db (2.9.1-1) ... \n\nProcessing triggers for ufw (0.36-0ubuntu1.1) ... \n
\n\n]
ubuntu@ubuntu:~$
```

```
ubuntu@ubuntu:~$ ansible local -i host.ini -m apt -a "name=nginx state=present update_cache=yes" --become
[DEPRECATION WARNING]: Distribution Ubuntu 20.04 on host localhost should use /usr/bin/python3, but is using /usr/bin/python for backward compatibility with prior Ansible releases. A future
Ansible release will default to using the discovered platform python for this host. See https://docs.ansible.com/ansible/2.9/reference_appendices/interpreter_discovery.html for more
information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
localhost | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python"
  },
  "cache_update_time": 1747566323,
  "cache_updated": true,
  "changed": true,
  "stderr": "",
  "stderr_lines": [],
  "stdout": "Reading package lists...Building dependency tree...Reading state information...The following packages were automatically installed and are no longer required:\n chromium
m-codecs-ffmpeg-extra gstreamer1.0-vaapi\n libgstreamer-plugins-bad1.0-0 libqt5concurrent5 libqt5opengl5-dev libqt5sql5\n libqt5sql5-sqlite libqt5test5 libvulkan-dev libwfreshark13 libwire
tap10\n libxext-dev qt5-gnome qt5-gnome-bins qtbase5-dev\n qtbase5-dev-tools x11proto-xext-dev\n sudo apt autoremove" to remove them.\n\nThe following additional packages will
be installed:\n libnginx-mod-http-image-filter libnginx-mod-http-xslt-filter\n libnginx-mod-mail libnginx-mod-stream nginx-common nginx-core\n\nSuggested packages:\n fcgiwrap nginx-doc\n\nThe following NEW packages will be installed:\n libnginx-mod-http-image-filter libnginx-mod-http-xslt-filter\n libnginx-mod-mail libnginx-mod-stream nginx-common nginx-core\n\n0 upgraded,
7 newly installed, 0 to remove and 67 not upgraded.\nNeed to get 605 kB of archives.\nAfter this operation, 2141 kB of additional disk space will be used.\n\nGet:1 http://ln.archive.ubuntu.c
om/ubuntu focal-updates/main amd64 nginx-common all 1.18.0-0ubuntu1.7 [37.8 kB]\nGet:2 http://ln.archive.ubuntu.com/ubuntu focal-updates/main amd64 libnginx-mod-http-image-filter amd64 1.18.
0-0ubuntu1.7 [14.8 kB]\nGet:3 http://ln.archive.ubuntu.com/ubuntu focal-updates/main amd64 libnginx-mod-http-xslt-filter amd64 1.18.0-0ubuntu1.7 [13.1 kB]\nGet:4 http://ln.archive.ubuntu.co
m/ubuntu focal-updates/main amd64 libnginx-mod-mail amd64 1.18.0-0ubuntu1.7 [43.0 kB]\nGet:5 http://ln.archive.ubuntu.com/ubuntu focal-updates/main amd64 libnginx-mod-stream amd64 1.18.0-bubu
ntui.7 [67.3 kB]\nGet:6 http://ln.archive.ubuntu.com/ubuntu focal-updates/main amd64 nginx-core amd64 1.18.0-0ubuntu1.7 [425 kB]\nGet:7 http://ln.archive.ubuntu.com/ubuntu focal-updates/main
amd64 nginx 1.18.0-0ubuntu1.7 [3620 B]\nPreconfiguring packages ...nfetched 605 kB in 7s (85.6 kB/s)\n\nSelecting previously unselected package nginx-common.\n\n(Reading database ... \n(
Reading database ... 55%\n(Reading database ... 10%\n(Reading database ... 15%\n(Reading database ... 20%\n(Reading database ... 25%\n(Reading database ... 30%\n(Reading database ... 35%\n(Rea
ding database ... 40%\n(Reading database ... 45%\n(Reading database ... 50%\n(Reading database ... 55%\n(Reading database ... 60%\n(Reading database ... 65%\n(Reading database ... 70%\n(Rea
ding database ... 75%\n(Reading database ... 80%\n(Reading database ... 85%\n(Reading database ... 90%\n(Reading database ... 95%\n(Reading database ... 100%\n(Reading database ... 222461 fi
les and directories currently installed.)\n\nPreparing to unpack .../0-nginx-common-1.18.0-0ubuntu1.7.all.deb ... \n\nUnpacking nginx-common (1.18.0-0ubuntu1.7) ... \n\nSelecting previously un
selected package libnginx-mod-http-image-filter.\n\nPreparing to unpack .../1-libnginx-mod-http-image-filter-1.18.0-0ubuntu1.7.and64.deb ... \n\nUnpacking libnginx-mod-http-xslt-filter (1.18.0-0bu
ntui.7) ... \n\nSelecting previously unselected package libnginx-mod-mail.\n\nPreparing to unpack .../3-libnginx-mod-mail-1.18.0-0ubuntu1.7.and64.deb ... \n\nUnpacking libnginx-mod-stream (1.18.0-0bu
ntui.7) ... \n\nSelecting previously unselected package nginx-core.\n\nPreparing to unpack .../5-nginx-core-1.18.0-0ubuntu1.7.and64.deb ... \n\nUnpacking nginx-core (1.18.0-0ubuntu1.7) ... \n
\nSelecting previously unselected package nginx.\n\nPreparing to unpack .../6-nginx-1.18.0-0ubuntu1.7.all.deb ... \n\nUnpacking nginx (1.18.0-0ubuntu1.7) ... \n\nSetting up nginx-common (1.18.0-0bu
ntui.7) ... \n\nCreated symlink /etc/systemd/system/multi-user.target.wants/nginx.service -> /lib/systemd/system/nginx.s
ervice.\n\nSetting up libnginx-mod-http-xslt-filter (1.18.0-0ubuntu1.7) ... \n\nSetting up libnginx-mod-mail (1.18.0-0ubuntu1.7) ... \n\nSetting up libnginx-mod-stream (1.18.0-0ubuntu1.7) ... \n
\nSetting up nginx-core (1.18.0-0ubuntu1.7) ... \n\nSetting up nginx (1.18.0-0ubuntu1.7) ... \n\nProcessing triggers for systemd (245.4-4ubuntu1.24) ... \n\nProcessing triggers for man-db (2.9.1-1) ... \n
\nProcessing triggers for ufw (0.36-0ubuntu1.1) ... \n\n\n]
  "stdout_lines": [
    "Reading package lists...",
    "Building dependency tree...",
    "Reading state information...",
    "The following packages were automatically installed and are no longer required:\n chromium
m-codecs-ffmpeg-extra gstreamer1.0-vaapi\n libgstreamer-plugins-bad1.0-0 libqt5concurrent5 libqt5opengl5-dev libqt5sql5\n libqt5sql5-sqlite libqt5test5 libvulkan-dev libwfreshark13 libwire
tap10\n libxext-dev qt5-gnome qt5-gnome-bins qtbase5-dev\n qtbase5-dev-tools x11proto-xext-dev\n sudo apt autoremove" to remove them.\n\nThe following additional packages will
be installed:\n libnginx-mod-http-image-filter libnginx-mod-http-xslt-filter\n libnginx-mod-mail libnginx-mod-stream nginx-common nginx-core\n\nSuggested packages:\n fcgiwrap nginx-doc\n\nThe following NEW packages will be installed:\n libnginx-mod-http-image-filter libnginx-mod-http-xslt-filter\n libnginx-mod-mail libnginx-mod-stream nginx-common nginx-core\n\n0 upgraded,
7 newly installed, 0 to remove and 67 not upgraded.\nNeed to get 605 kB of archives.\nAfter this operation, 2141 kB of additional disk space will be used.\n\nGet:1 http://ln.archive.ubuntu.c
om/ubuntu focal-updates/main amd64 nginx-common all 1.18.0-0ubuntu1.7 [37.8 kB]\nGet:2 http://ln.archive.ubuntu.com/ubuntu focal-updates/main amd64 libnginx-mod-http-image-filter amd64 1.18.
0-0ubuntu1.7 [14.8 kB]\nGet:3 http://ln.archive.ubuntu.com/ubuntu focal-updates/main amd64 libnginx-mod-http-xslt-filter amd64 1.18.0-0ubuntu1.7 [13.1 kB]\nGet:4 http://ln.archive.ubuntu.co
m/ubuntu focal-updates/main amd64 libnginx-mod-mail amd64 1.18.0-0ubuntu1.7 [43.0 kB]\nGet:5 http://ln.archive.ubuntu.com/ubuntu focal-updates/main amd64 libnginx-mod-stream amd64 1.18.0-bubu
ntui.7 [67.3 kB]\nGet:6 http://ln.archive.ubuntu.com/ubuntu focal-updates/main amd64 nginx-core amd64 1.18.0-0ubuntu1.7 [425 kB]\nGet:7 http://ln.archive.ubuntu.com/ubuntu focal-updates/main
amd64 nginx 1.18.0-0ubuntu1.7 [3620 B]\nPreconfiguring packages ...nfetched 605 kB in 7s (85.6 kB/s)\n\nSelecting previously unselected package nginx-common.\n\n(Reading database ... \n(
Reading database ... 55%\n(Reading database ... 10%\n(Reading database ... 15%\n(Reading database ... 20%\n(Reading database ... 25%\n(Reading database ... 30%\n(Reading database ... 35%\n(Rea
ding database ... 40%\n(Reading database ... 45%\n(Reading database ... 50%\n(Reading database ... 55%\n(Reading database ... 60%\n(Reading database ... 65%\n(Reading database ... 70%\n(Rea
ding database ... 75%\n(Reading database ... 80%\n(Reading database ... 85%\n(Reading database ... 90%\n(Reading database ... 95%\n(Reading database ... 100%\n(Reading database ... 222461 fi
les and directories currently installed.)\n\nPreparing to unpack .../0-nginx-common-1.18.0-0ubuntu1.7.all.deb ... \n\nUnpacking nginx-common (1.18.0-0ubuntu1.7) ... \n\nSelecting previously un
selected package libnginx-mod-http-image-filter.\n\nPreparing to unpack .../1-libnginx-mod-http-image-filter-1.18.0-0ubuntu1.7.and64.deb ... \n\nUnpacking libnginx-mod-http-xslt-filter (1.18.0-0bu
ntui.7) ... \n\nSelecting previously unselected package libnginx-mod-mail.\n\nPreparing to unpack .../3-libnginx-mod-mail-1.18.0-0ubuntu1.7.and64.deb ... \n\nUnpacking libnginx-mod-stream (1.18.0-0bu
ntui.7) ... \n\nSelecting previously unselected package nginx-core.\n\nPreparing to unpack .../5-nginx-core-1.18.0-0ubuntu1.7.and64.deb ... \n\nUnpacking nginx-core (1.18.0-0ubuntu1.7) ... \n
\nSelecting previously unselected package nginx.\n\nPreparing to unpack .../6-nginx-1.18.0-0ubuntu1.7.all.deb ... \n\nUnpacking nginx (1.18.0-0ubuntu1.7) ... \n\nSetting up nginx-common (1.18.0-0bu
ntui.7) ... \n\nCreated symlink /etc/systemd/system/multi-user.target.wants/nginx.service -> /lib/systemd/system/nginx.s
ervice.\n\nSetting up libnginx-mod-http-xslt-filter (1.18.0-0ubuntu1.7) ... \n\nSetting up libnginx-mod-mail (1.18.0-0ubuntu1.7) ... \n\nSetting up libnginx-mod-stream (1.18.0-0ubuntu1.7) ... \n
\nSetting up nginx-core (1.18.0-0ubuntu1.7) ... \n\nSetting up nginx (1.18.0-0ubuntu1.7) ... \n\nProcessing triggers for systemd (245.4-4ubuntu1.24) ... \n\nProcessing triggers for man-db (2.9.1-1) ... \n
\nProcessing triggers for ufw (0.36-0ubuntu1.1) ... \n\n\n]
  ]
}
```

4. Start a service

ansible local -i host.ini -m service -a "name=nginx state=started" --become

```
ubuntu@ubuntu:~$ ansible local -i host.ini -m service -a "name=nginx state=started" --become
[DEPRECATION WARNING]: Distribution ubuntu 20.04 on host localhost should use /usr/bin/python3, but is using /usr/bin/python for backward compatibility with prior Ansible releases. A future
Ansible release will default to using the discovered platform python for this host. See https://docs.ansible.com/ansible/2.9/reference_appendices/interpreter_discovery.html for more
information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
localhost | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python"
  },
  "changed": false,
  "name": "nginx",
  "state": "started",
  "status": {
    "ActiveEnterTimestamp": "Sun 2025-05-18 16:35:34 IST",
    "ActiveEnterTimestampMonotonic": "8907189075",
    "ActiveExitTimestampMonotonic": "0",
    "ActiveState": "active",
    "After": "basic.target systemd-journald.socket network.target system.slice sysinit.target",
    "AllowIsolate": "no",
    "AllowedCPUs": "",
    "AllowedMemoryNodes": "",
    "AmbientCapabilities": "",
    "AssertResult": "yes",
    "AssertTimestamp": "Sun 2025-05-18 16:35:34 IST",
    "AssertTimestampMonotonic": "8907060424",
    "Before": "multi-user.target shutdown.target",
    "BlockIOAccounting": "no",
    "BlockIOWeight": "[not set]",
    "CPUAccounting": "no",
    "CPUAffinity": "",
    "CPUAffinityFromNUMA": "no",
    "CPUQuotaPerSecUseC": "infinity",
    "CPUQuotaPeriodUseC": "infinity",
    "CPUSchedulingPolicy": "0",
    "CPUSchedulingPriority": "0",
    "CPUSchedulingResetOnFork": "no",
    "CPUShares": "[not set]",
    "CPUUsageSec": "[not set]",
    "CPUWeight": "[not set]",
    "CacheDirectoryMode": "0755",
    "CanIsolate": "no",
    "CanReload": "yes",
    "CanStart": "yes",
    "UnitFileState": "enabled",
    "UtmpMode": "init",
    "WantedBy": "multi-user.target",
    "WatchdogSignal": "6",
    "WatchdogTimestampMonotonic": "0",
    "WatchdogUsec": "0"
  }
}
ubuntu@ubuntu:~$
```

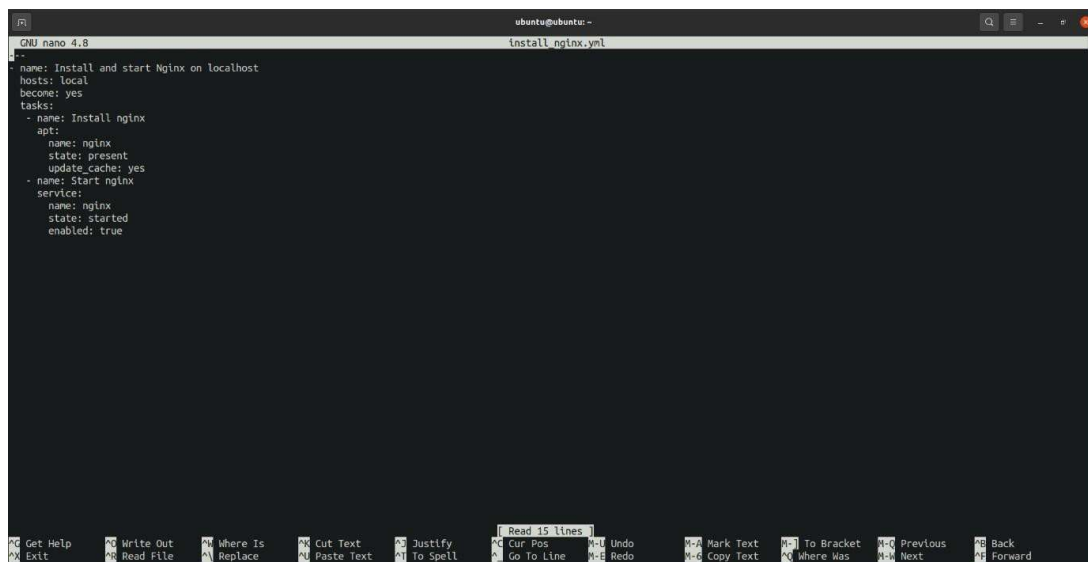
Practical No.:12

Demonstrate Ansible Playbooks

Install and Start Nginx

install_nginx.yml:

```
- name: Install and start Nginx on web servers
hosts: webservers
become: true
tasks:
  - name: Install Nginx
    apt:
      name: nginx
      state: present
      update_cache: yes
  - name: Start Nginx
    service:
      name: nginx
      state: started
      enabled: true
```



The screenshot shows a terminal window titled 'ubuntu@ubuntu: ~' with a nano editor open to a file named 'install_nginx.yml'. The file content is the same Ansible playbook as shown in the text block above. The terminal window has a dark background and a light-colored text. The nano editor's status bar at the bottom shows 'GNU nano 4.8' and 'install_nginx.yml'. The bottom of the terminal window displays a row of keyboard shortcuts for nano, such as 'Get Help', 'Write Out', 'Where Is', 'Cut Text', 'Justify', 'Cur Pos', 'Undo', 'Mark Text', 'To Bracket', 'Previous', 'Back', 'Exit', 'Read File', 'Replace', 'Paste Text', 'To Spell', 'Go To Line', 'Redo', 'Copy Text', 'Where Was', 'Next', and 'Forward'.

MCA SEM II 2024-25

DES's NMITD
DevOps Lab

Roll no. C24120

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DEVOPS LAB
C24079

ROLL NO:

Run the Playbook:

ansible-playbook -i hosts.ini install_nginx.yml

```
ubuntu@ubuntu:~$ ansible-playbook -i host.ini install_nginx.yml

PLAY [Install and start Nginx on localhost] *****

TASK [Gathering Facts] *****
[DEPRECATION WARNING]: Distribution Ubuntu 20.04 on host localhost should use /usr/bin/python3, but is using /usr/bin/python for backward compatibility with prior Ansible releases. A future
Ansible release will default to using the discovered platform python for this host. See https://docs.ansible.com/ansible/2.9/reference_appendices/interpreter_discovery.html for more
information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
ok: [localhost]

TASK [Install nginx] *****
ok: [localhost]

TASK [Start nginx] *****
ok: [localhost]

PLAY RECAP *****
localhost : ok=3  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
```

